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系统枢纽
SYSTEMS HUB

智能交通学域
INTELLIGENT TRANSPORTATION THRUST

End-to-End Navigation II

Graduate Course INTR-6000P

Week 12 - Lecture 24

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End-to-End Self-Driving

2014–2018: Modular + HD map

2019–2021: Sensor fusion + planning

2022–2023: BEV + multi-task learning

2024→ : End-to-end Navigation + VLM grounding

End-to-End Self-Driving

Pipeline Overview:

Sensors → BEV → Perception → Behavior prediction → Trajectory → Control

Modules include:
•BEV mapping
•Object & lane detection
•Agent intent prediction
•Trajectory head
•Safety controller

Choice	Options
Input	Camera / LiDAR / IMU / GPS
Representation	Raw pixels / BEV / voxel grid / 4D attention
Output	Control / trajectory / affordances
Training	IL, RL, self-supervision
World Model	Transformer temporal memory, diffusion models

End-to-End Self-Driving

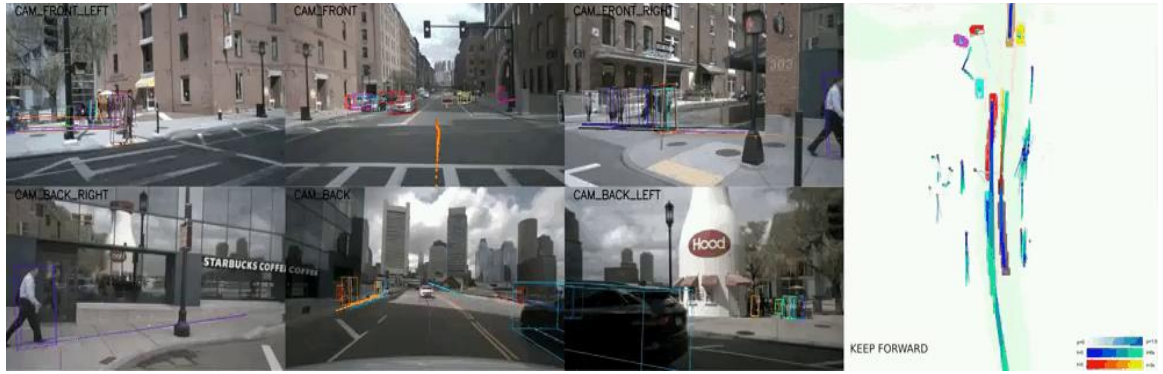
Why End-to-End for Driving?

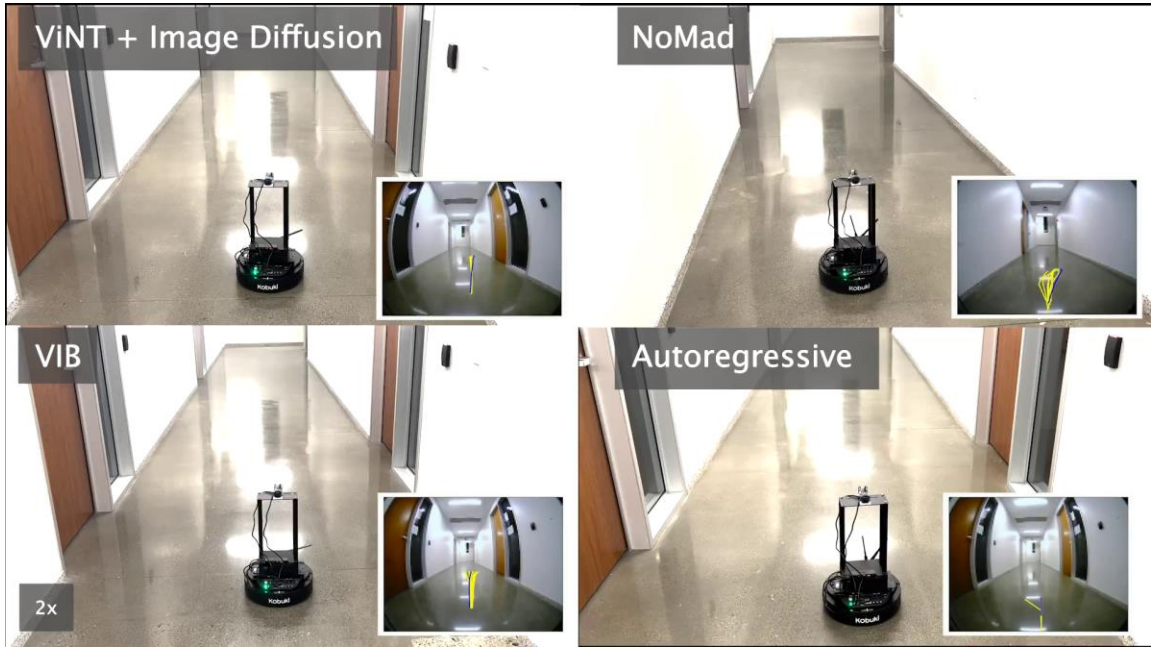
Conventional stack issues:

- Manual feature engineering brittle
- HD maps expensive & degrade
- Strong human priors limit generalization

End-to-end advantages:

- Joint learning = fewer failure cascades
- Learning from real driving behavior
- Better long-tail behavior & emergent priors





Typical End-to-End Self-Driving Models

Direct Control Prediction

Core Concept: A single deep learning model maps raw sensor data (e.g., images) directly to vehicle control commands (steering, throttle, brake).

Exemplar Models: PilotNet (NVidia), CILRS (Wayve), LAV (UC Berkeley).

Key Advantage: Ultra-fast inference, enabling real-time reaction.

Primary Challenge: Limited interpretability, creating a "black box" problem for validation and debugging.

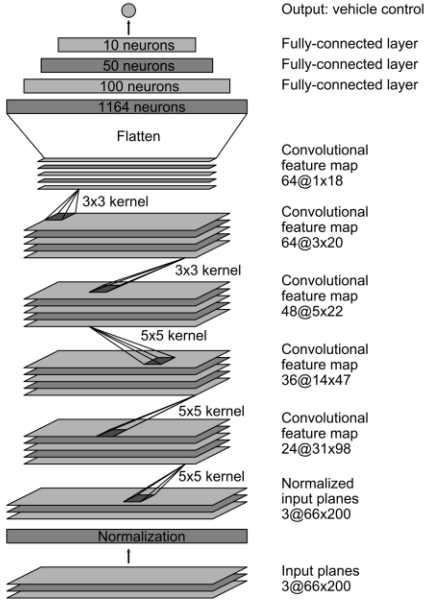
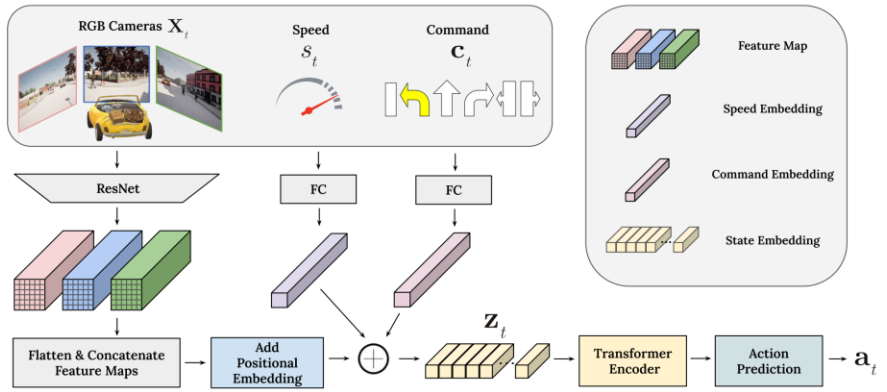
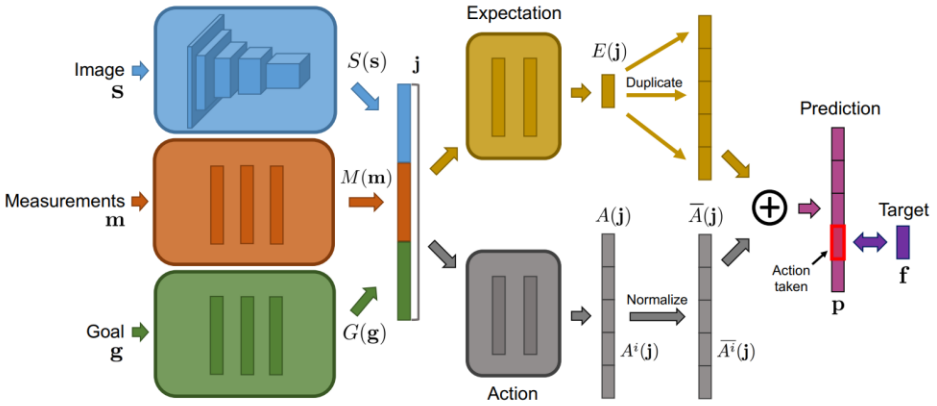


Figure 1: PilotNet architecture.

PilotNet



CILRS



LAV

Typical End-to-End Self-Driving Models

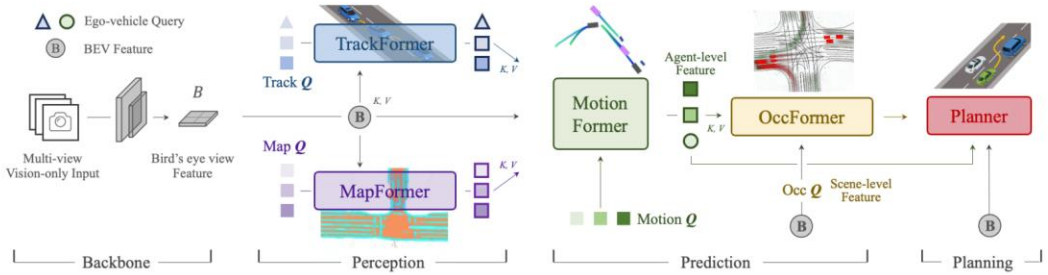
Intermediate Representation-Driven Models

Exemplar Models: UniAD, VAD, TCP

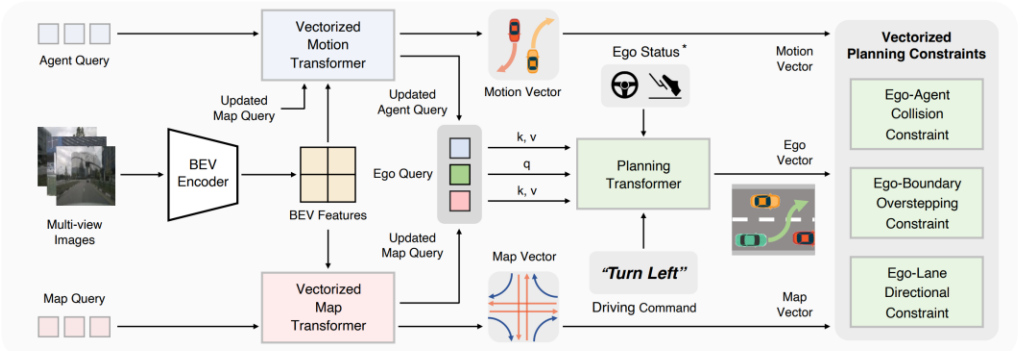
Core Innovation: Learns a rich, internal Bird's-Eye View (BEV) or semantic map as a foundational world model.

Structured Workflow:

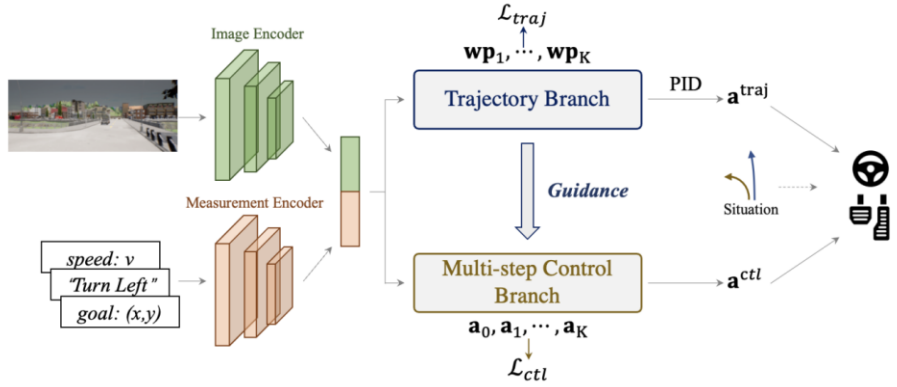
- Perception: Fuses sensor data to construct a dynamic BEV map, encoding the geometry and semantics of the scene.
- Planning: Uses this rich representation to predict others' trajectories and plan a safe, comfortable, and compliant ego-path.
- Control: Executes the planned trajectory with smooth and precise vehicle control.



UniAD



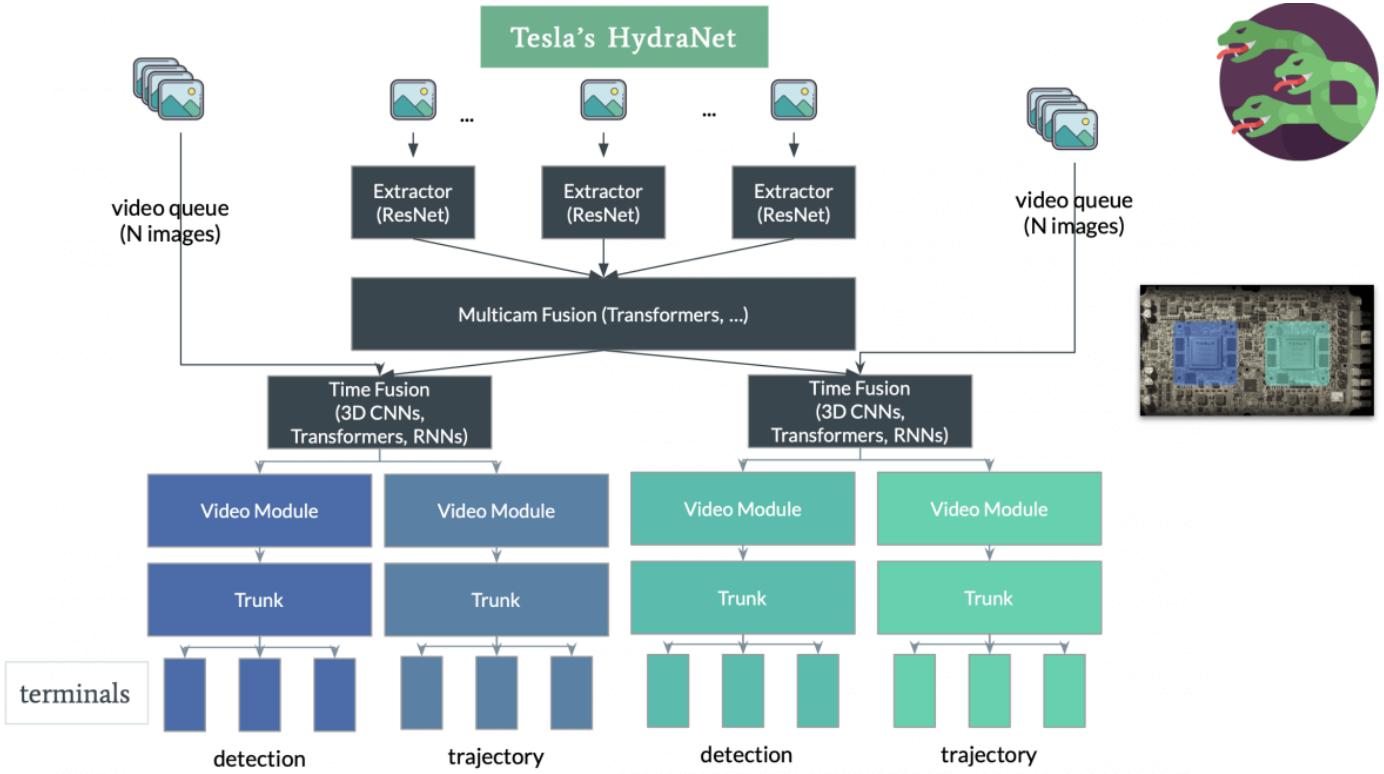
VAD



TCPNet

Case Study: Tesla FSD (Vision-only)

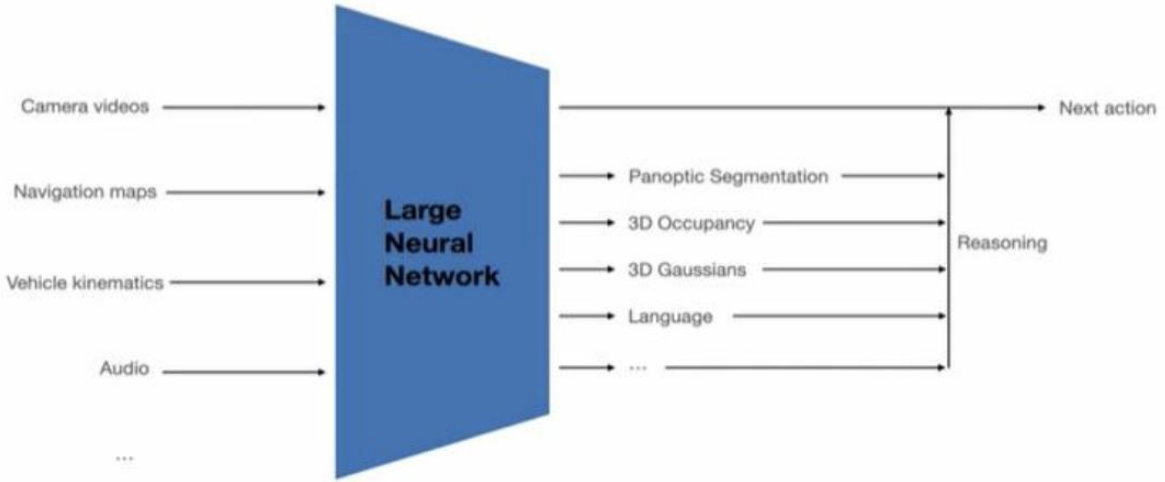
Tesla FSD: Fully end-to-end systems (2021):
"It's like Chat-GPT, but for cars!!!" "Instead of determining the proper path of the car based on rules, we determine the car's proper path by using neural network that learns from millions of training examples of what humans have done."



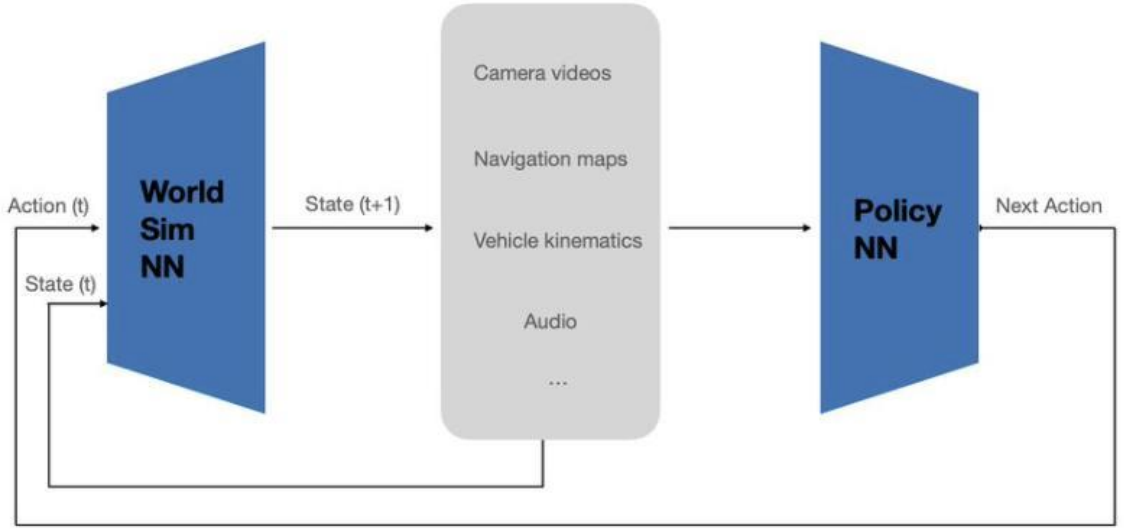
Case Study: Tesla FSD (Vision-only)

Tesla FSD: Fully end-to-end systems (2025)

Main Challenges of learning pixels -> control
1. Curse of dimensionality
2.Interpretability and safety guarantees
3. Evaluation



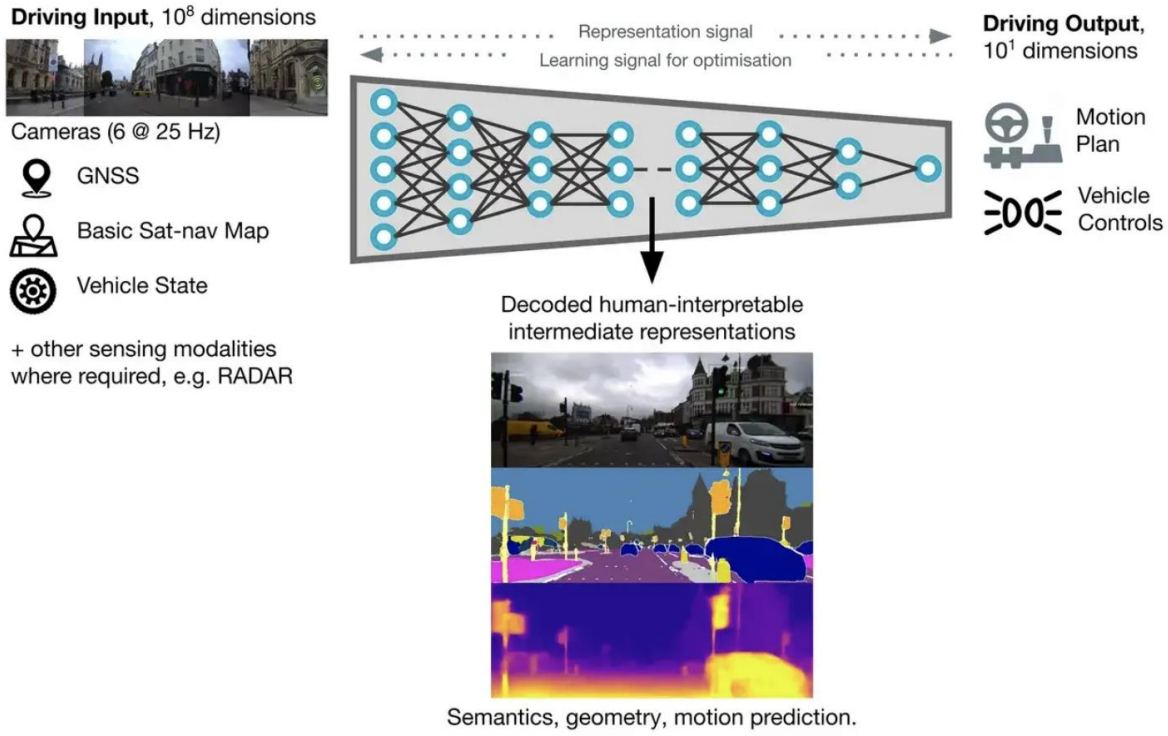
Interpretability and safety guarantees

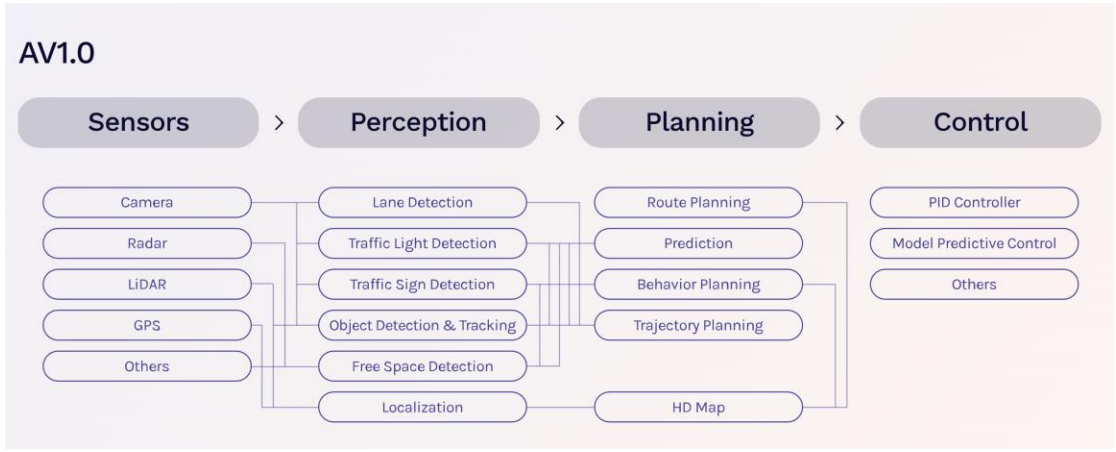


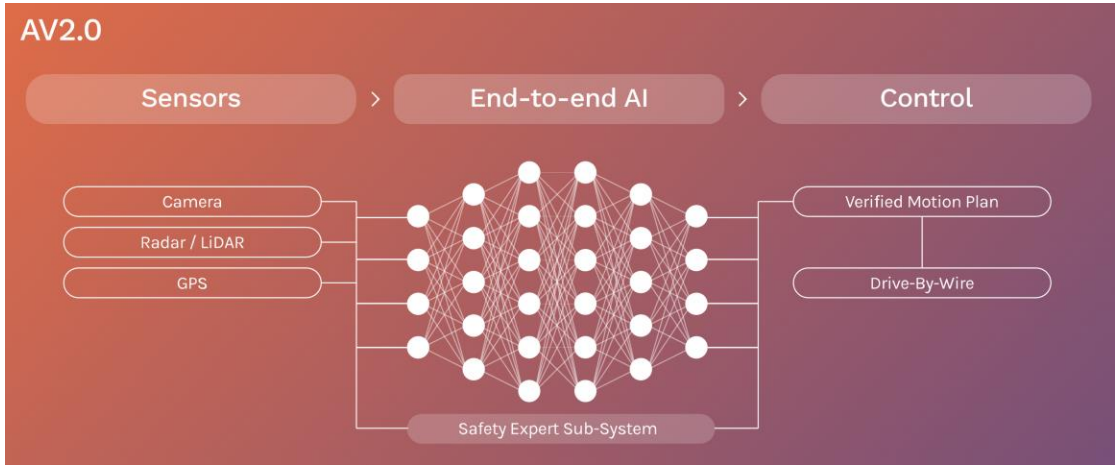
Neural Network closed-loop simulatorCan be trained with cheap to collect state-action data

Case Study: Wayve

Our innovative approach replaces the modular ‘sense-plan-act’ architecture of the traditional AV1.0 approach with a single neural network trained on diverse data to convert raw sensor inputs into safe driving outputs.







Safety & Reliability

Key challenges:

- Corner cases
- Uncertainty calibration
- Formal safety layers
- Rule grounding & driver norms
- Adversarial robustness
- Regulatory requirements

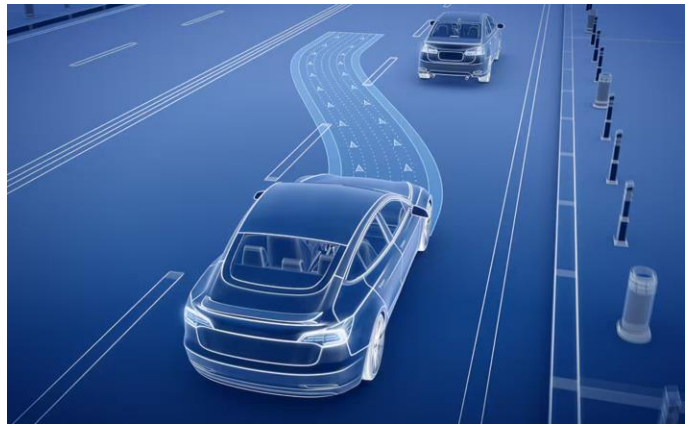
Takeaways:

- E2E does not mean “black-box”
- BEV + temporal transformers dominate
- Industry uses hybrid safety stack
- Learned planning → improved generalization

Embodied Navigation Beyond Cars

Platforms:

- Indoor service robots
- Warehouse AMRs
- Delivery robots (sidewalk bots)
- Drones (inspection, racing)
- Home humanoids



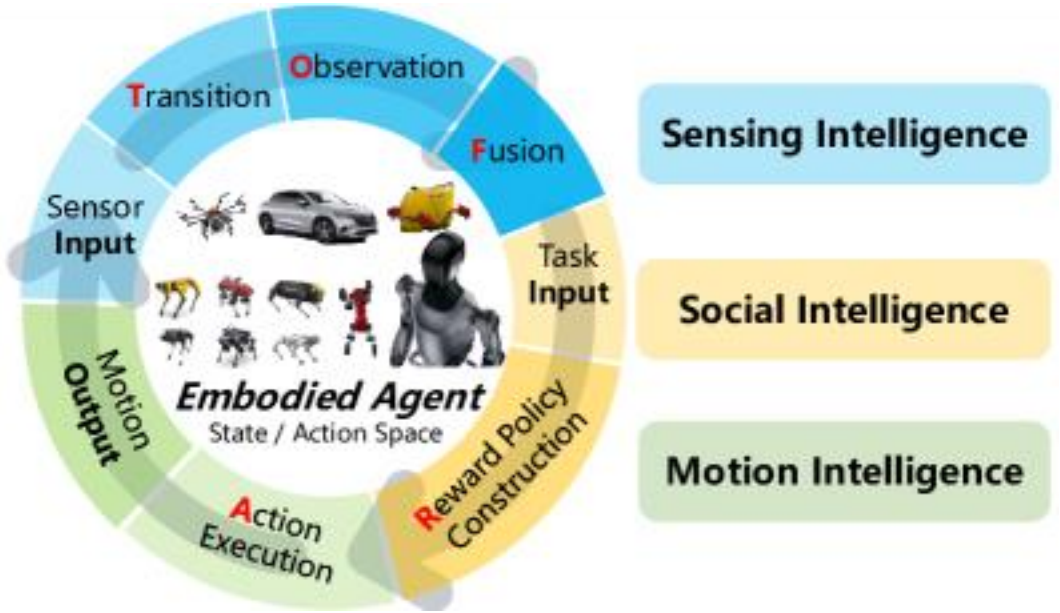
Embodied Navigation Loop

Pipeline:

Observe → Understand → Navigate → Interact

Sensors:

- RGB, depth, stereo
- LiDAR, IMU, wheel odometry
- Language instructions (VLM era)
- Tactile for manipulation robots



Source: Sensing, Social, and Motion Intelligence in Embodied Navigation: A Comprehensive Survey

End-to-End Robot Navigation

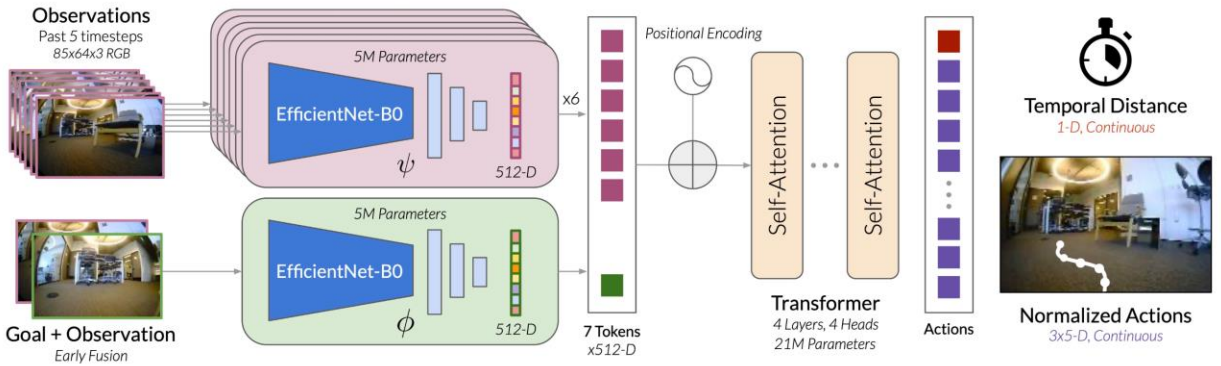
Vision based Robot Navigation Model:

ViNT: A foundation model for visual navigation

Architecture: Transformer-based, pre-trained on large-scale, heterogeneous robot data.

Mechanism: Goal-conditioned sequence prediction. Encodes a history of images and a goal image to autoregressively predict velocity commands.

Strength: Demonstrates strong few-shot adaptation and generalization to new robots and environments.



D. Shah, A. Sridhar, N. Dashora, K. Stachowicz, K. Black, N. Hirose, and S. Levine, "ViNT: A foundation model for visual navigation," in 7th Annual Conference on Robot Learning, 2023. [Online]. Available: https://arxiv.org/abs/2306.14846

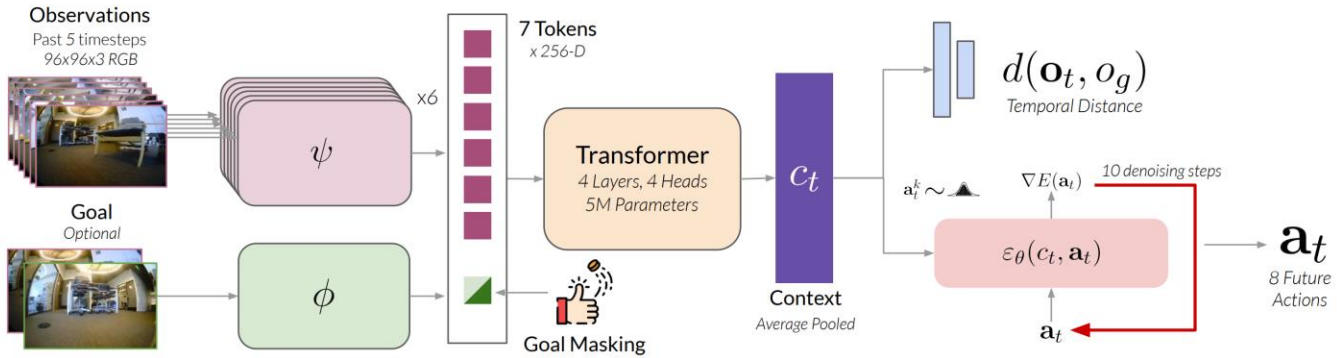
End-to-End Robot Navigation

Vision based Robot Navigation Model:

NoMaD: Goal Masked Diffusion Policies for Navigation and Exploration

Imitation learning model that uses diffusion policies to generate robust and diverse navigation behaviors.

Core Idea: Employs a masked goal-conditioning strategy and normalized data to handle multi-modal scenarios and improve generalization.



A. Sridhar, D. Shah, C. Glossop, and S. Levine, "NoMaD: Goal Masked Diffusion Policies for Navigation and Exploration," arXiv pre-print, 2023. [Online]. Available: https://arxiv.org/abs/2310.07896

End-to-End Robot Navigation

Vision-Language-Action (VLA) Agents

Agents that map:

(Images,Language)→Navigation+Manipulation

Examples:
- SayCan / RT-1 / RT-2
- PaLM-E
- MobileVLA / Embodied-GPT

End-to-End Robot Navigation

Vision-Language Navigation (VLN)

Task examples:
- “Go to the kitchen and find the refrigerator.”
- “Navigate to the red sofa and open the drawer.”

Benchmarks:
- R2R, RxR
- ALFRED / Ego-VQA
- Habitat-VLN
- BEHAVIOR-1K

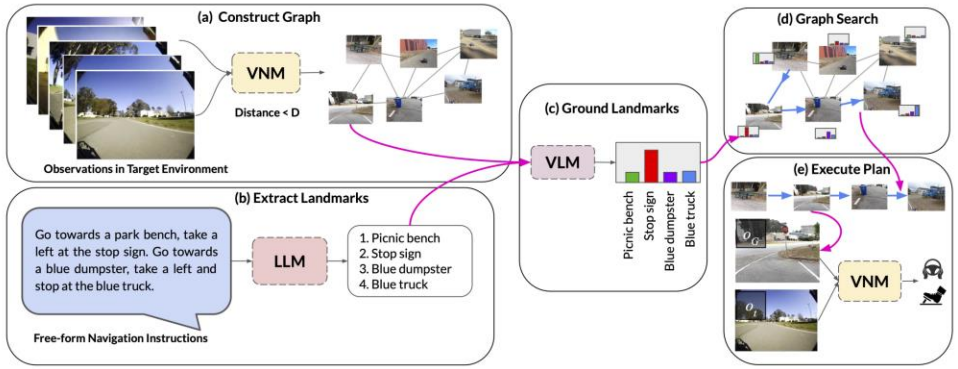
End-to-End Robot Navigation

Typical VLN Model:

LM-Nav:

Concept: Uses a large language model (LLM) as a semantic planner. It translates high-level natural language commands ("get my coffee from the kitchen") into a sequence of visual landmarks to follow.

Flow: Language → LLM → Landmark Sequence ("go to the hallway, then the kitchen") → Visuomotor Policy → Actions.



D. Shah, B. Osinski, B. Ichter, and S. Levine, "LM-nav: Robotic navigation with large pre-trained models of language, vision, and action," in 6th Annual Conference on Robot Learning, 2022. [Online]. Available: https://openreview.net/forum?id=UW5A3SweAH

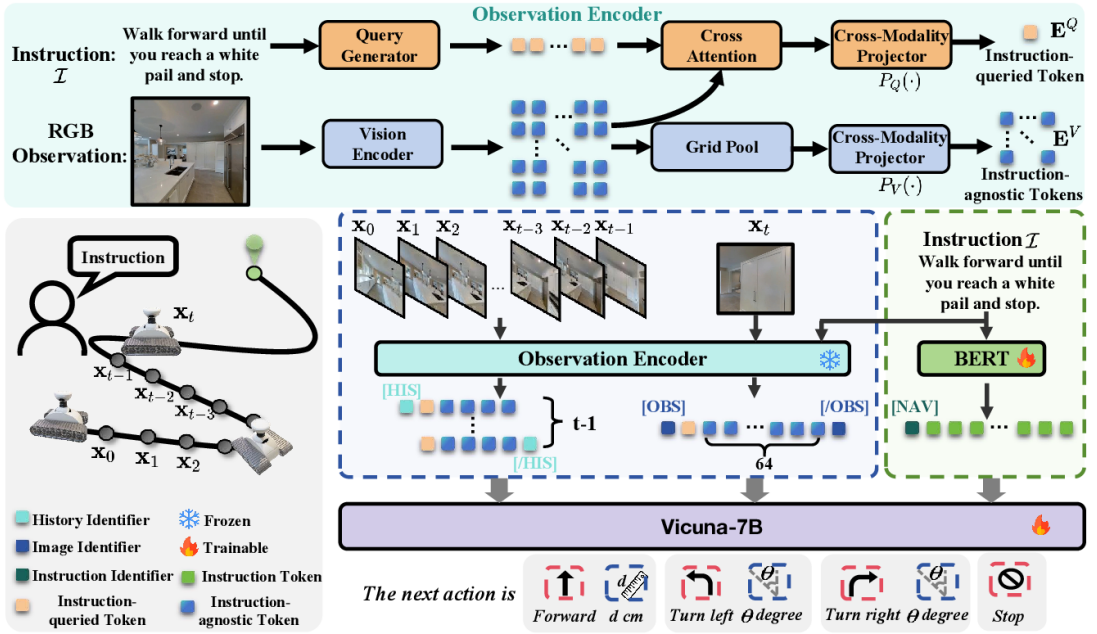
End-to-End Robot Navigation

Typical VLN Model:

Navid:

Concept: An LLM-powered multimodal agent that controls a robot directly from instructions and visual input. It reasons about actions in real-time.

Flow: Continuously integrates language instruction, camera images, and its own actions to decide the next move, enabling closed-loop control.



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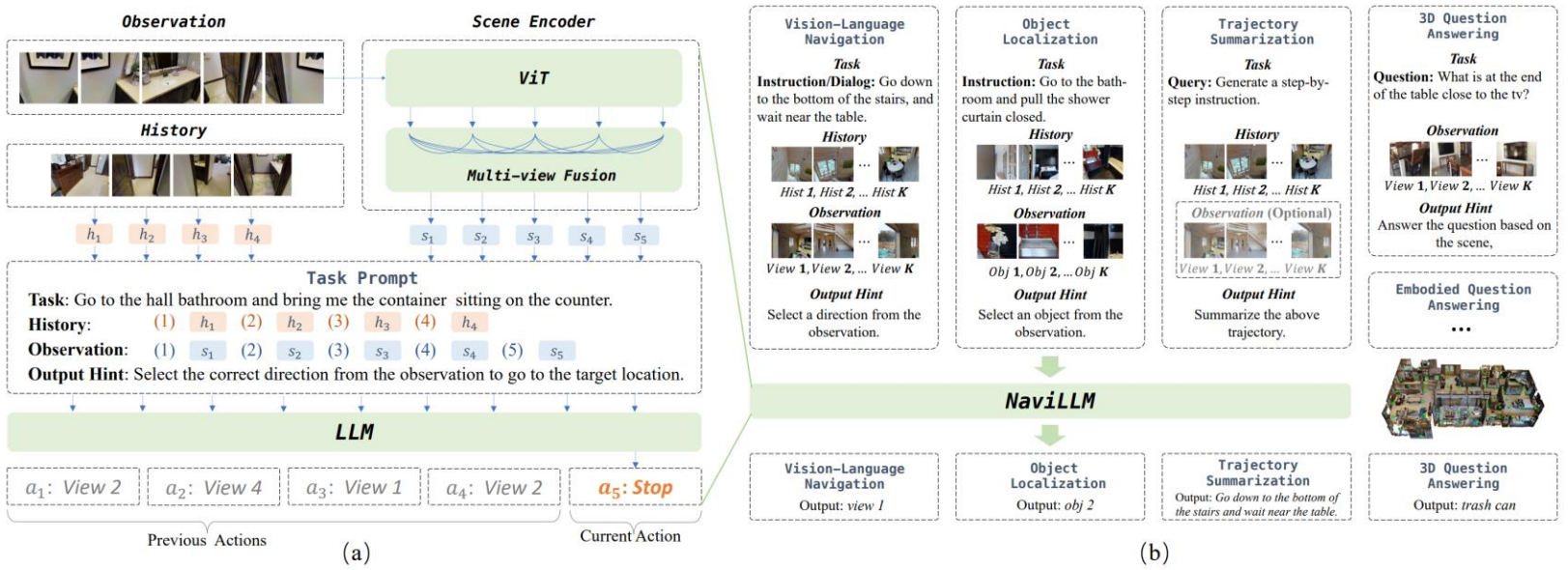
End-to-End Robot Navigation

Typical VLN Model:

GPT-Navigator:

Extend Large Language Models (LLMs) from simple text processing to embodied AI applications that can understand and navigate the physical world.



Source: D. Zheng, S. Huang, L. Zhao, Y. Zhong, and L. Wang, “Towards learning a generalist model for embodied navigation,” in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 13 624–13 634.

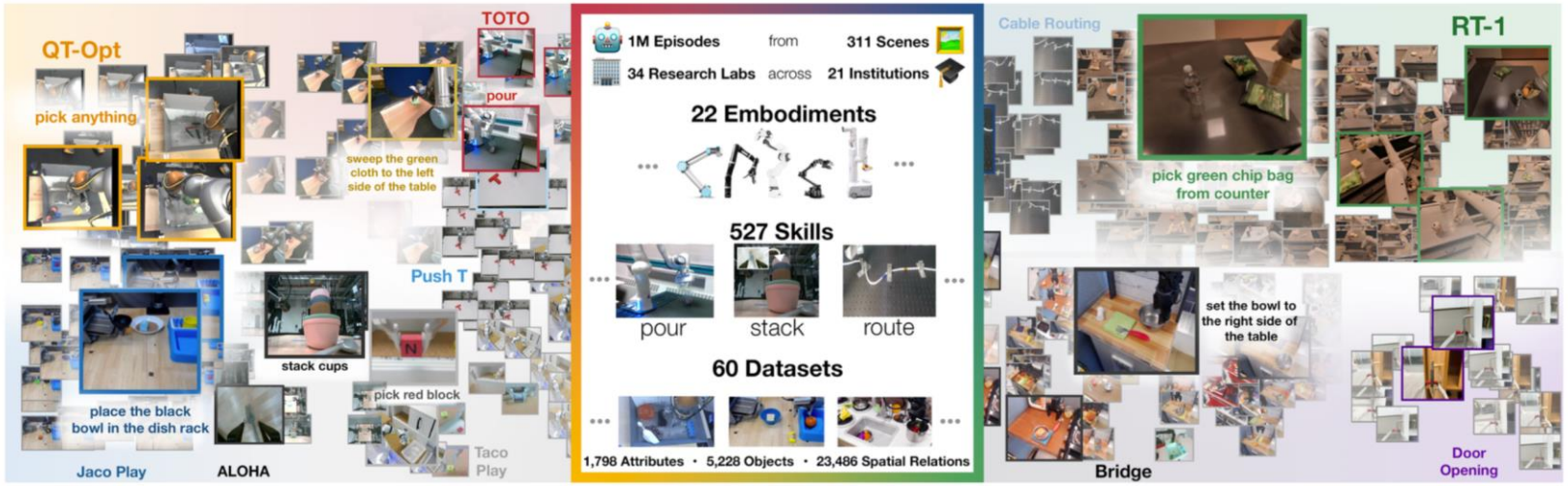
Generalist Robot Models

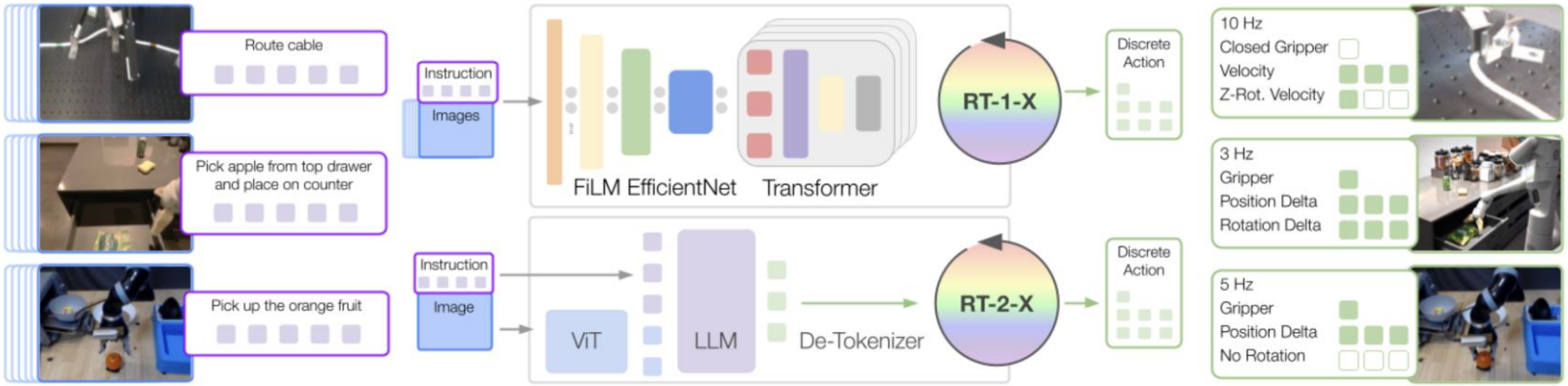
Trends:

- Single model → many robots/tasks
- Action tokens for robots
- Diffusion-based trajectory planners
- LLM as policy & skill routing

Examples:

- Open-X Embodiment
- RT-X





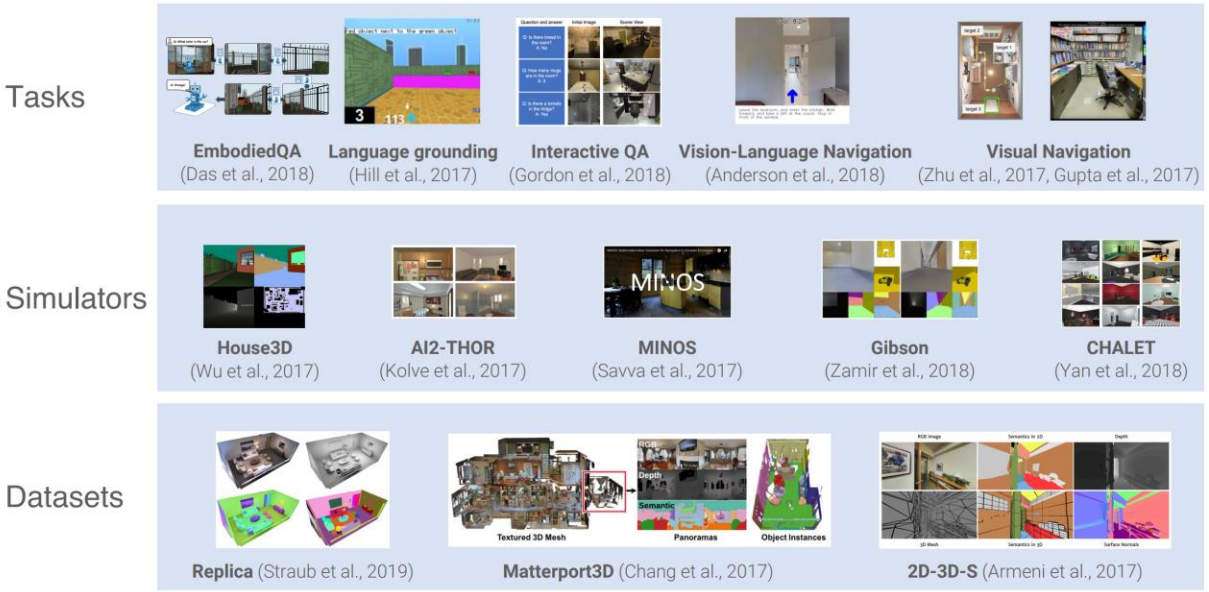
Simulation Platforms

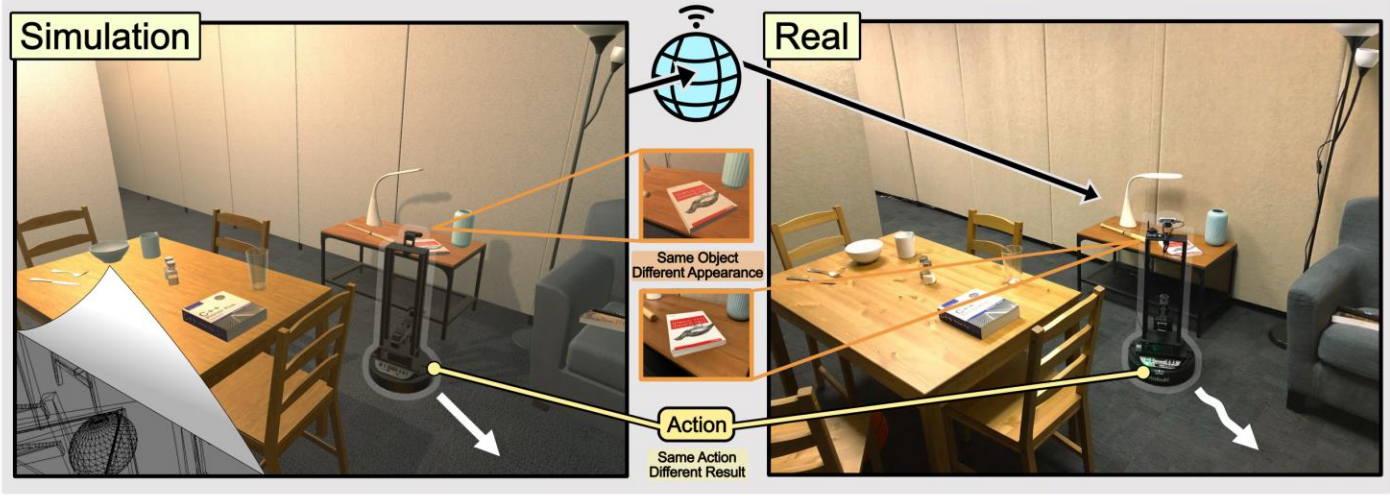
Habitat:

A flexible, high-performance 3D simulator with configurable agents, multiple sensors, and generic 3D dataset handling

RoboTHOR:

RoboTHOR is an environment within the AI2-THOR framework, designed to develop embodied AI agents. It consists of a series of scenes in simulation with counterparts in the physical world.





Simulation Platforms

Isaac Sim:

NVIDIA Isaac Sim™ is an open-source reference application built on NVIDIA Omniverse™ that enables developers to simulate and test AI-driven robotics solutions in physically based virtual environments.

Learning Paradigms for Navigation Models

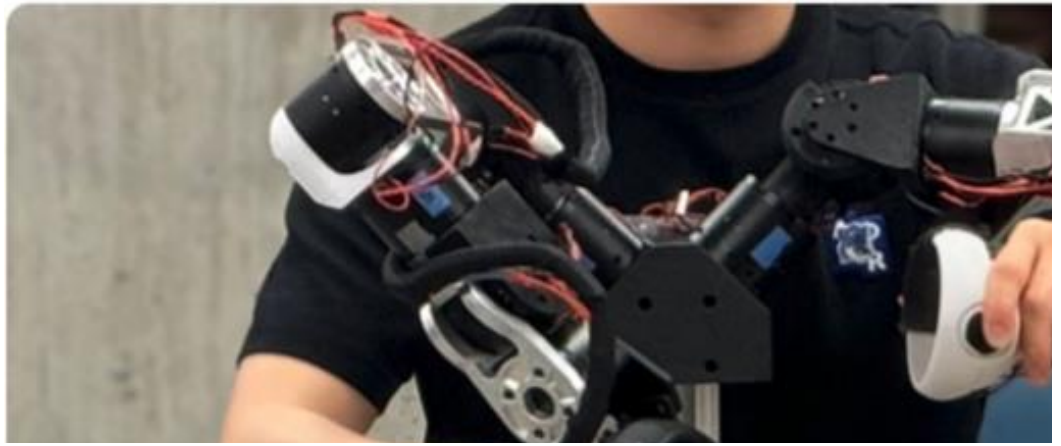
Imitation Learning

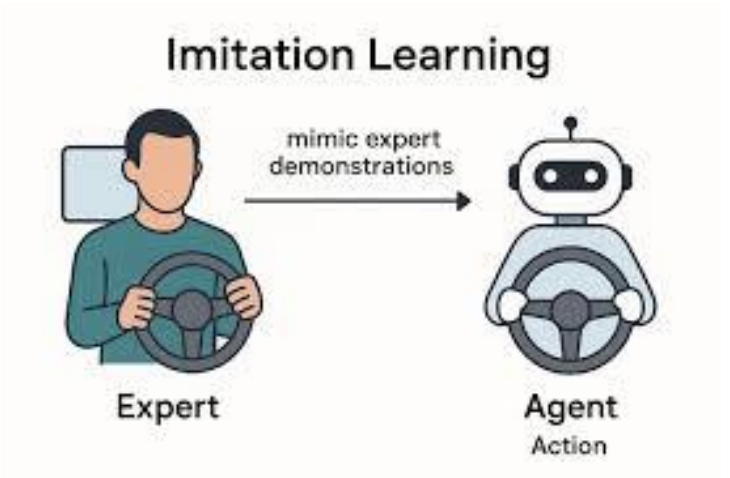
Behavioral Cloning (BC): Mimics an expert's state-action mapping via supervised learning.

Dataset Aggregation (DAgger): Iteratively corrects BC's errors by querying the expert for labels on the agent's own visited states.

Inverse Reinforcement Learning (IRL): Infers the hidden reward function that explains the expert's behavior, rather than just mimicking actions.

Adversarial Imitation Learning (AIL): A discriminator network judges if a trajectory is from the expert or the agent, training the agent to deceive the discriminator.

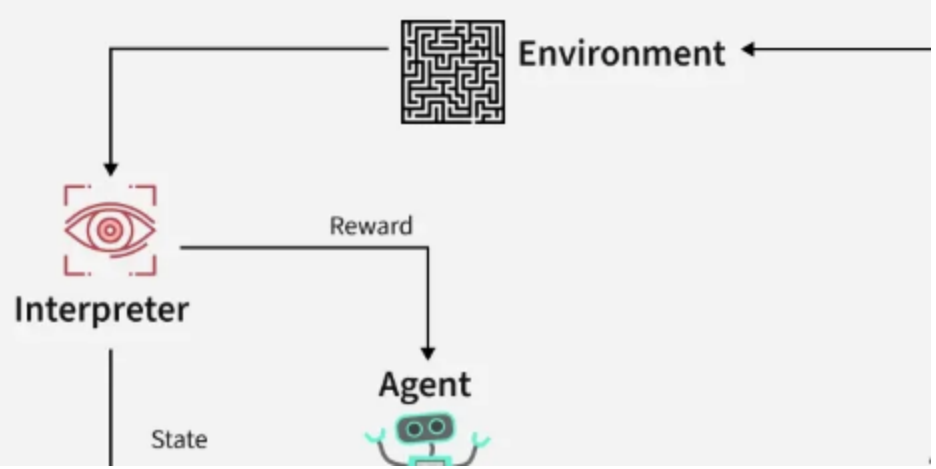




Learning Paradigms for Navigation Models

Reinforcement Learning:

Core Idea: An agent learns optimal actions through trial and error in an environment.



Mechanism: The agent maximizes a cumulative reward signal by discovering which behaviors yield the best long-term outcomes.

Analogy: Learning to navigate by exploration and receiving feedback (rewards/punishments).

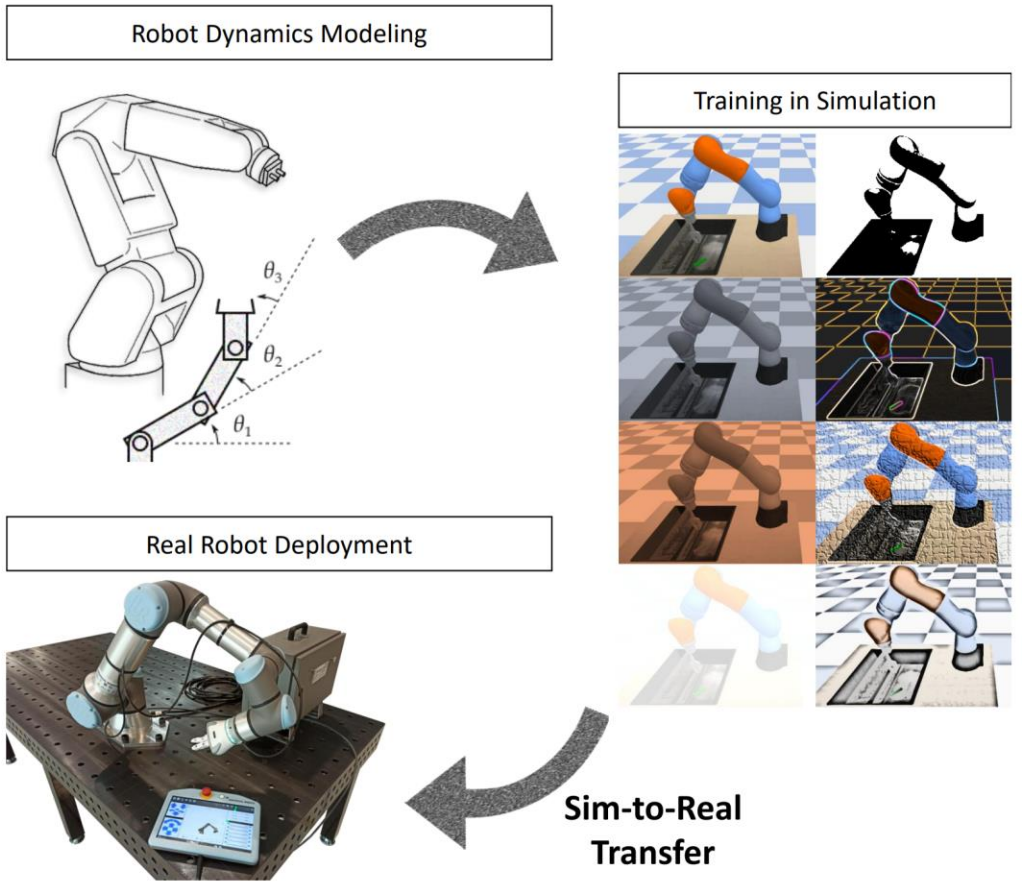
Learning Paradigms for Navigation Models

Transfer Learning

Core Idea: Leverage pre-existing knowledge from a source task to accelerate learning and improve performance on a new, target task.

Benefit: Avoids "starting from scratch," saving significant compute resources and data requirements.

Analogy: A professional racer using their skills to quickly master a new type of vehicle.



New Trends and Discussion

Open Questions

1. The Architectural Dilemma

What is the optimal balance between interpretable modular systems and end-to-end models?

2. The Scalability vs. Efficiency Trade-off

How can we develop end-to-end models that are both large enough to be capable and small enough for real-time deployment?

3. Guaranteeing Safety & Alignment

How do we provably integrate hard safety constraints and ethical rules into data-driven learning systems?

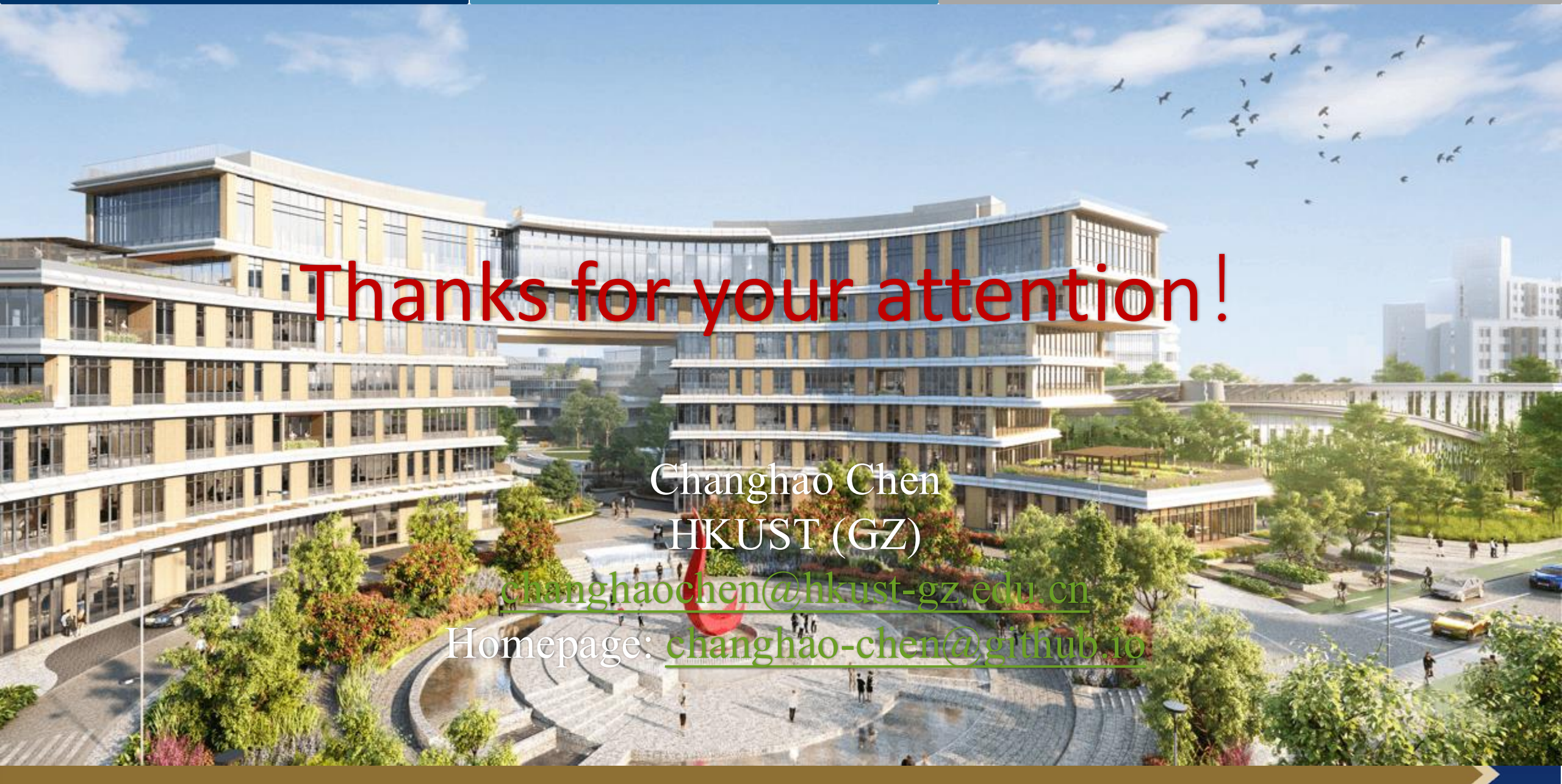
New Trends and Discussion

Key Challenges in Embodied AI

- Sim-to-real transfer
- Long-horizon memory
- 3D reasoning + object permanence
- Social navigation
- Safety + human comfort
- Efficient data collection

Summary

- Self-driving → hybrid E2E stacks w/ BEV transformers
- Embodied AI → multimodal learning (vision-language-action)
- World models & robot foundation models emerging
- Safety, reliability, grounding remain hardest problems



Thanks for your attention!

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